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MIL-L-24296A(SHIPS)
3 August 1970
SUPERSEDING
MIL-L-24296(SHIPS)
15 August 1968
(See 6.3)

MILITARY SPECIFICATION

LOUDSPEAKER, PRESSURE PROOF, TYPE LS-450( )/B

1. SCOPE

1.1 Scope. This specification covers a pressure proof loudspeaker for use as a microphone and a loudspeaker in escape hatches and on the bridge of a submarine.

1.2 Classification. Loudspeakers furnished under this specification shall be type LS-450( )/B. The type designation contains parentheses which are to be replaced by a letter indicating a particular make or model that differs from previous production. Nomenclature shall be requested as required by MIL-E-21981.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.
MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.
MIL-E-21981 - Electronics Type Designations, Identification Plates and Markings: Requirements for.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

DRAWING

MILITARY

815-1853087 - Loudspeaker, Pressure Proof, Type LS-450( )/B.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
S1.10-1966 - Calibration of Microphones, Methods for the.
S1.12-1967 - Laboratory Standard Microphones, Specifications for.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
219 - Loudspeaker Measurements, Recommended Practice For.

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, N.Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Loudspeakers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.2).

3.2 Description. The LS-450( )/B loudspeaker shall be a permanent magnet cone type loudspeaker designed for use aboard Navy submarines in exterior locations such as escape hatches or the bridge. The loudspeaker shall be capable of operating both as a loudspeaker and as a microphone to provide for two-way voice communication. The loudspeaker design shall provide for free access of water or air to both sides of the cone to enable the loudspeaker to withstand the required hydrostatic pressure and gunblast tests. The voice coil shall be encapsulated in a silicone rubber form which completely fills the magnetic air gap and protects it against entry of water and foreign matter.

3.3 General features. The loudspeaker shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein (see 6.1). Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.

General requirements
Substitution of (equal or superior) materials or parts
Painting
Protection against corrosion
Locking devices
Threaded devices
Rounded corners and edges
Drilled and tapped holes
Structural welding
Gunblast (see 3.6)
Shock, vibration, and inclination (see 3.6)
Dielectric strength and insulation resistance clearances (see 3.6)
Standardization
Repair parts (electronic)
Workmanship

3.4 Mechanical requirements. The loudspeaker shall be constructed in accordance with Drawing 815-1853087. It shall be capable of withstanding the required environmental tests and of meeting the performance requirements specified herein.

3.4.1 Size and mounting dimensions. The loudspeaker size and mounting dimensions shall conform to Drawing 815-1853087.

3.4.2 Weight. Maximum weight of the loudspeaker shall be 15 pounds.

3.5 Electrical and performance requirements.

3.5.1 Impedance. The loudspeaker voice coil impedance shall be 16 $\pm$ 2 ohms at a frequency of 1000 hertz (Hz).

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3.5.2 Quality of reproduction. There shall be no buzzes, rattles or other spurious sounds arising from a malfunction of the loudspeaker when tested in accordance with 4.5.1.

3.5.3 Maximum power input. The maximum electrical power input for the loudspeaker shall not exceed 5 volt-amperes when measured in accordance with 4.5.3.

3.5.4 Sound pressure output. When tested in accordance with 4.5.4.1, the loudspeaker shall be capable of producing a minimum sound pressure level of 96 decibels (db.)[1/] for the 700-1300 Hz warble band measured on axis at a distance of 5 feet in front of the loudspeaker. Following the environmental tests specified in 4.5.6 through 4.5.14 the measured sound pressure output shall be within 2 db. of the original value measured before the environmental tests.

3.5.5 Frequency response. The frequency response, measured as specified in 4.5.4.2, shall be within the limits shown on figure 1.

3.5.6 Load characteristic. The load characteristic (sound pressure output versus volt-ampere input) of the loudspeaker shall be essentially linear. The maximum deviation of the load curve from true linearity (3 db. increase in sound pressure output for twice the volt-ampere input) shall be 0.5 db. when measured as specified in 4.5.4.3.

3.5.7 Sensitivity as a microphone. The output voltage developed by the loudspeaker under the test conditions specified in 4.5.4.4 shall be equal to or greater than 2 millivolts.

3.5.8 Voice coil polarity. One of the loudspeaker terminals shall be marked "+". Application of a d.c. voltage to the loudspeaker terminals with positive voltage to the "+" terminal shall result in the cone moving outward.

3.6 Operation under conditions of shipboard environment. The loudspeaker shall show no evidence of physical damage and shall be capable of meeting the performance requirements specified in 3.5.2, 3.5.4, 3.5.5, and 3.5.7 after being subjected to the environmental tests specified in 4.5.6 through 4.5.14. Following each of the specified environmental tests the quality of reproduction, sound pressure output, frequency response and sensitivity as a microphone tests shall be performed to determine compliance with 3.5.2, 3.5.4, 3.5.5, and 3.5.7.

3.7 Instruction sheets. One set of instruction sheets shall be furnished with each loudspeaker. The instruction sheets shall be marked "NAVSHIPS 0965-042-9010" for identification purposes and shall be approved at the time of qualification approval by the procuring activity having technical cognizance. Instruction sheets shall include the following:

(a) Brief description of the loudspeaker.
(b) Installation instructions for the loudspeaker including size and location of mounting holes.
(c) Loudspeaker performance and electrical data.
(d) Federal stock number for loudspeaker cone and procedure for replacing the cone.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests.[2/] Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification inspection shall consist of the examination and tests specified in table I.

1/ Relative to 0.0002 microbars (dynes per square centimeter).

2/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.2 and 6.2.1).

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Table I - Qualification tests.

Examination and tests	Requirement paragraph	Examination and test reference
General examination	3.3	MIL-I-983
Dielectric strength	3.3	4.5.2
Insulation resistance	3.3	MIL-I-983
Quality of reproduction	3.5.2	4.5.1
Power input	3.5.3	4.5.3
Sound pressure output	3.5.4	4.5.4.1
Frequency response	3.5.5	4.5.4.2
Load characteristic	3.5.6	4.5.4.3
Sensitivity as a microphone	3.5.7	4.5.4.4
Polarity	3.5.8	4.5.5
Endurance	3.6	4.5.6
Hydrostatic pressure	3.6	4.5.7
Fungus	3.6	4.5.8
Salt spray	3.6	4.5.9
Freezing	3.6	4.5.10
Extreme cold temperature	3.6	4.5.11
Gunblast	3.3 and 3.6	4.5.12
Vibration	3.3 and 3.6	4.5.13
Shock	3.3 and 3.6	4.5.14

4.3 Sampling for quality conformance inspection.

4.3.1 Inspection lot. All equipment of the same type presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.3.2 Sampling for group A examination and tests. A sample of equipments, shall be selected from each inspection lot in accordance with table II and subjected to the group A examination and tests listed in table IV. The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A examination and test shall be counted as a defect, and the equipment shall not be offered for delivery. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by that sample.

Table II - Sampling for group A examination and tests.

Number of equipments in inspection lot	Number of equipments in sample	Number of equipments nonconforming on group A examination and tests	
		Acceptance number	Rejection number
7 and under	All	-	-
8 to 15	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

4.3.3 Sampling for group B tests. A sample of equipments shall be selected from each inspection lot in accordance with table III and subjected to the group B tests listed in table IV. The results of each test shall be compared with specification requirements. Failure to conform to the specification requirements for any group B test shall be counted as a defect, and the equipment shall not be offered for delivery. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by that sample.

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Table III - Sampling for group B tests.

Number of equipments in inspection lot	Number of equipments in sample	Number of equipments nonconforming on group B tests	
		Acceptance number	Rejection number
3 and under	All	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.3.4 Sampling for group C tests. Group C tests will be required by the procuring activity when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to the group C tests listed in table IV. The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.4 Quality conformance inspection. The sample equipments selected in accordance with 4.3 shall be subjected to the examination and tests listed in table IV. Tests shall be performed, in general, in the order listed.

Table IV - Quality conformance inspection.

Examination and tests	Requirement paragraph	Examination and test reference
Group A		
General examination	3.3	MIL-I-983
Quality of reproduction	3.5.2	4.5.1
Sound pressure output	3.5.4	4.5.4.1
Group B		
Load characteristic	3.5.6	4.5.4.3
Frequency response	3.5.5	4.5.4.2
Power input	3.5.3	4.5.3
Sensitivity as a microphone	3.5.7	4.5.4.4
Polarity	3.5.8	4.5.5
Insulation resistance	3.3	MIL-I-983
Group C		
Dielectric strength	3.3	4.5.2
Endurance	3.6	4.5.6
Hydrostatic pressure	3.6	4.5.7
Fungus	3.6	4.5.8
Salt spray	3.6	4.5.9
Freezing	3.6	4.5.10
Extreme cold temperature	3.6	4.5.11
Gunblast	3.3 and 3.6	4.5.12
Vibration	3.3 and 3.6	4.5.13
Shock	3.3 and 3.6	4.5.14

4.5 Test procedures.

4.5.1 Quality of reproduction. A 9 volt root mean square (r.m.s.) signal shall be applied across the input terminals of the loudspeaker. The frequency of the input signal shall be varied from 600 to 5000 Hz and the acoustic output checked for compliance with 3.5.2.

4.5.2 Dielectric strength. Dielectric strength tests shall be conducted as specified in MIL-I-983. The voice coil circuit shall be tested while a signal is being applied as specified in 4.5.4. The test voltage for the voice coil circuit shall be 300 volts r.m.s. applied between one voice coil terminal and the surrounding metal parts.

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4.5.3 Power input. A 9 volt r.m.s. signal warbled between the limits of 700 to 1300 Hz shall be applied to the input terminals. The warble signal shall be such that the logarithm of frequency changes linearly with time. The frequency variation from the lower limit to the upper limit shall occur at a rate of 5 to 6 Hz. The current drawn by the loudspeaker shall be measured to determine compliance with 3.5.3.

4.5.4 Acoustical performance tests. Loudspeaker acoustical performance tests specified in 4.5.4.1 through 4.5.4.4 shall be conducted with the loudspeaker positioned in a free field acoustic environment as described in IEEE Number 219. The ambient temperature in the sound room shall be between 20° and 30°C. The microphone used for measuring the sound pressure shall conform to ANSI S1.10-1966 and be calibrated by the method described therein. Other types of microphones may be used if calibrated as described in ANSI S1.12-1967. The microphone shall be positioned with its axis colinear with the axis of the loudspeaker cone and a distance of 5 feet between the microphone face and loudspeaker cover.

4.5.4.1 Sound pressure output. A 9 volt r.m.s. warble signal as specified in 4.5.3 shall be applied to the input terminals of the loudspeaker. The r.m.s. value of sound pressure output shall be measured to determine compliance with 3.5.4.

4.5.4.2 Frequency response. A 9 volt r.m.s. signal shall be applied to the loudspeaker input terminals. The signal frequency shall be varied continuously from 600 to 5000 Hz. The sound pressure output shall be measured to determine compliance with 3.5.5.

4.5.4.3 Load characteristic. With a warble signal 700 to 1300 Hz the input voltage shall be varied from 0.5 to 9 volts in not less than eight steps (approximately input 3 db. steps) and the sound pressure level and input current measured at each step. The sound pressure level in db. shall be plotted against input in volt-amperes on semi-log paper. The resulting curve shall meet the requirements of 3.5.6.

4.5.4.4 Sensitivity as a microphone. The loudspeaker voice coil terminals shall be connected across a 16 ohm non-inductive resistor. A 700-1300 Hz warble signal at 4 microbars r.m.s. sound pressure level (measured at the face of the loudspeaker) shall be applied. The voltage developed across the resistor shall be measured to determine compliance with 3.5.7.

4.5.5 Polarity. A 6 volt d.c. potential shall be applied across the loudspeaker terminals with the positive terminal of the source connected to the "+" terminal of the loudspeaker. The loudspeaker cone shall move away from the magnetic structure at the moment of application in accordance with 3.5.8.

4.5.6 Endurance. A warble frequency signal of 700 to 1300 Hz at 9 volts r.m.s. shall be impressed across the input terminals of the loudspeaker for a continuous period of 100 hours. The loudspeaker shall then be operated for one additional hour with the input voltage raised to 13 volts r.m.s. The ambient temperature of air around the loudspeaker shall be 130°F. throughout this test. Loudspeaker performance following the endurance test shall be in accordance with 3.6.

4.5.7 Hydrostatic pressure. The loudspeaker shall be tested at a pressure of 694 pounds per square inch (p.s.i.) in accordance with the submersible (694 p.s.i.) test specified in MIL-STD-108. Following the pressure test the loudspeaker performance shall be checked for conformance with 3.6.

4.5.8 Fungus. The loudspeaker shall be subjected to the fungus resistance test in accordance with MIL-E-5272. The speaker shall then be tested for conformance with 3.6.

4.5.9 Salt spray. The loudspeaker shall be subjected to a salt spray test performed in accordance with method 101 of MIL-STD-202. The salt solution concentration shall be 20 percent. Length of the test shall be 96 hours. Following the salt spray test the loudspeaker shall conform to 3.6.

4.5.10 Freezing. The loudspeaker shall be immersed in water 2°C. above freezing and left there to stabilize for 30 minutes. The speaker shall be removed from the water into an atmosphere of minus 35°C. After 10 minutes the performance of the speaker shall be tested for conformance with 3.6.

4.5.11 Extreme cold temperature. The loudspeaker shall be subjected to the minus 40°F. temperature extreme of the extended storage test specified in MIL-I-983. After stabilization to room temperature the loudspeaker shall be tested for compliance with 3.6.

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4.5.12 Gunblast. The loudspeaker shall be subjected to the simulated gunblast test specified in MIL-I-983.

4.5.13 Vibration. The loudspeaker shall be mounted on a simulated bulkhead and subjected to the vibration test for vital equipment specified in MIL-I-983.

4.5.14 Shock. The loudspeaker shall be mounted on a simulated bulkhead and subjected to the shock test specified in MIL-I-983.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

5.2 Domestic shipment and early equipment installation and for storage of onboard repair parts (see 5.4.1).

5.2.1 Loudspeakers.

5.2.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.2.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2.2 Onboard repair parts. Onboard repair parts shall be preserved and packaged level A, packed level C and marked levels A and C respectively, in accordance with MIL-E-17555.

5.3 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 5.4.2 and 6.1).

5.3.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.3.1.1 Preservation, packaging, packing and marking. The equipment shall be preserved and packaged levels A or C; packed level A or B as specified and marked in accordance with MIL-E-17555.

5.4 Use of polystyrene (loose-fill) material.

5.4.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be

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marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.4.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of items required.
(c) Inventory Control Point for repair parts (see 3.3).
(d) Quantity of repair parts (see 3.3).
(e) Level of preservation, packaging, packing and marking required if other than as specified in 5.2 (see 5.3).
(f) When polystyrene (loose-fill) material is approved (see 5.4.1).

6.2 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-24296 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.2.1).

6.2.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
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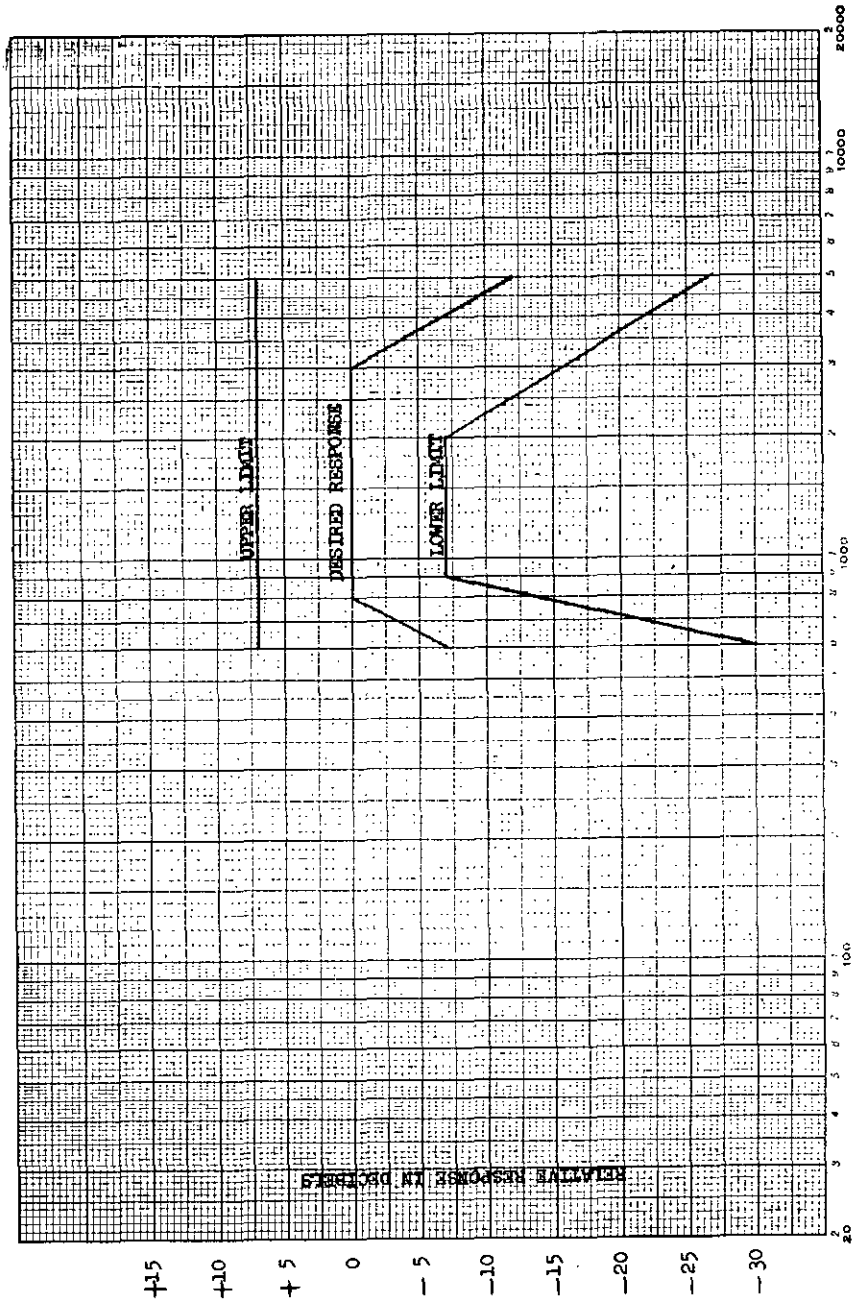


FIGURE 1 FREQUENCY RESPONSE LIMITS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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Naval Ship Engineering Center
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Prince George's Center
Hyattsville, Maryland 20782

OFFICIAL BUSINESS

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Commander, Naval Ship Engineering Center
DOD Standardization Program & Documents Branch
Center Building
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